Characteristics of Flow and Flame Behavior Behind Rifled/Unrifled Nozzles

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A novel rifled nozzle was installed behind a conventional combustion exhauster to improve combustion efficiency. The rifled nozzles improve the momentum transmission, turbulent strength, and mixing efficiency between the central jet and annular jet. The flow characteristics behind the nozzles (rifled and unrifled) were visualized and detected using the smoke-wire flow visualization, particle image velocimetry, and hot-wire anemometry. The cold flow structures were categorized into four modes—jet flow, single bubble, dual bubble, and turbulent flow. The topological scheme was adopted to analyze and verify these flow modes. The flame structures behind the nozzles (rifled and unrifled) are classified into three modes—jet flame, flickering flame, and turbulent flame—using the direct-photo visualization. The flame height of a 12-rifled nozzle is decreased by about 50% under that of an unrifled nozzle. The flame shedding frequency declines rapidly in the flickering flame mode and the relationship between the Strouhal number (Sr) and annular velocity (u_a) is $Sr = 0.0238 + 0.13/u_a$. [DOI: 10.1115/1.3078388]

Keywords: rifled nozzles, flow characteristics, flame behavior

1 Introduction

Bluff bodies and swirling flow were utilized to increase the mixing between air and fuel flows and to improve the efficiency of combustion and heat transfer in industrial combustors, flow mixers, and other industrial equipments. The effects of swirling flow and bluff body form a low-pressure recirculation zone near the nozzle outlet and then increase the diffusion between the central jet-flow and recirculation zone. The increase in turbulent intensity and interaction time causes an intense mixing between the central jet and the recirculation zones. Accordingly, the mixing concentration and the combustion stability are increased. Gupta et al. [1], Rose [2], Chigier and Beer [3], and Davies and Beer [4] examined the typical flow properties in the recirculation zone such as the velocity profile and pressure distribution, and determined the relationship between the combustion efficiency and the bluff body/swirling jet.

Numerous recent studies have examined the flow around a bluff body flame stabilizer in both reacting and nonreacting environments. Carmody [5] used a pitot tube to delineate the mean flow field and pressure distribution when a free annular jet-flow moved around a circular bluff disk. Carmody found that 95% of energy in the mean flow was converted to an intensive turbulent disturbance within the range of $3r$, where r is the disk radius. Davies and Beer [4] utilized the hot-wire anemometry to investigate the mean flow structures, the effect of blockage ratio, and the turbulent properties when a free annular jet flew through a bluff body. They also defined three flow regions—recirculation region, transition region, and established region where the flow behavior in the established region is similar to that in the jet flow mode. Durao and Whitelaw [6] measured the mean velocity field, turbulent properties, probability density function (pdf) distributions, and energy spectra using a laser Doppler velocimeter (LDV) when a free annular jet flow passed a bluff body. Durao and Whitelaw indicated that the turbulent flow is not isotropic and that a larger blockage ratio of bluff body was associated with a shorter recirculation zone. Schefer et al. [7] examined the flow structures, velocity profiles, and the axial pdf-distribution between central-penetrating and unpenetrating flows using the two-dimensional LDV. Schefer et al. [7] found a dual-peak pdf-distribution near the rim of the central jet flow and a Gaussian pdf-distribution in the other regions.

Huang and Lin [8] investigated the flame temperature and the velocity distribution of a circular bluff disk (blockage ratio =0.44) by using the direct-photo and Schlieren flow visualization techniques. Huang and Lin classified the flame structures into seven modes—recirculated flame, transition flame, unsteady detached flame, laminar ring flame, developing flame, split flashing flame, and lift flame. They also found that a triple flame existed at the rim of the blue annular flame in the laminar ring-flame mode. Gu et al. [9] studied the unsteady aerodynamic structures and the stability limit of the turbulent nonpremixed flame using the humidity effect and the particle image velocimetry (PIV) scheme. Gu et al. [9] categorized the flame into four modes—recirculation zone flame, transition flame, central-jet dominated flame, and partially quenched flame—using different velocities of air and fuel. Gu et al. [9] determined that the increase in combustion reacting time significantly reduced the limit of the partially quenching flame when humid air joined the flow.

Many researchers have examined various mechanisms to produce swirling flow. Rose [2], Hartnett and Eckert [10], and Hübner et al. [11] mobilized axial swirling flow by rotating a cylinder. The cylinder was rotated from rest to 5000 rev/min (rpm) with an increment of 500 rpm, and the swirling flow was driven by the tangential linear momentum that was generated by the viscosity. However, the swirl intensity is weak because the viscosity of air is low. Chigier and Beer [3], Cheng [12], and Al-Abdeli and Masri [13] stirred the swirling flow by conducting a tangential flow into a swirl generator. Al-Abdeli and Masri [13] utilized a tangential inlet flow to control the swirling flow and increase the swirling number (S) as high as 1.59. Kerr and Fraser [14], Guttenfelder et al. [15], and Huang and Tsai [16] excited the swirling flow by changing the angle of a swirl vane pack. Huang and Tsai applied a swirl vane pack with an angle-tunable wing-airfoil to generate swirling flow. They found that the global unsteady flow field and the increase in turbulent intensity dominate the flow mixing.

Chervinsky [17] utilized the theoretical analysis and experimental results to examine the width, length, and temperature of

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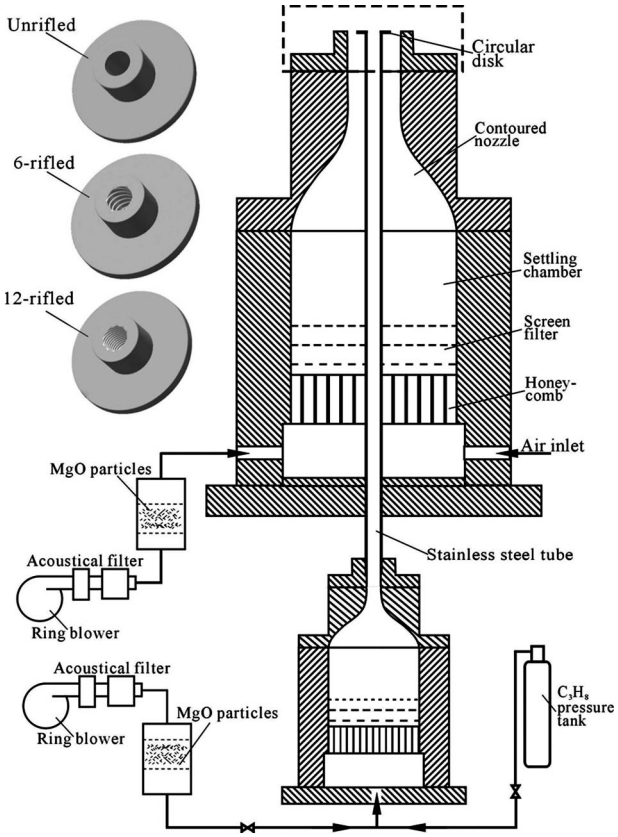

Fig. 1 Experimental setup

flames and the mass entrainment at a fixed flux of C_3H_8 with various swirling numbers. Chervinsky found that the increase in swirling number increases the flame width, the highest temperature, and the mass entrainment; however, it decreases the flame height. Chen and Driscoll [18] investigated the flame height of swirling flow and the velocity profile in the recirculation using the flow observation scheme. Chen and Driscoll [18] found that the recirculation increases the mixing efficiency of a large-scale flow structure. They also found that the increase in swirling number shortens the flame height and velocity in the recirculation zone, and that the flame height approaches a fixed value as $S>0.5$. Morcos and Abdel-Rahim [19] observed the flame length in both nonswirling and swirling flows. They found that the flame length generally decreased with increasing air-fuel mass flow rate ratio, burner tube diameter, axial distance between combustor exit and entrance of burner tube, fuel-air pressure ratio, and degree of swirling represented by air tangential angular speed. However, the flame length increased with increasing burner tube length. Morcos and Abdel-Rahim [19] also claimed that the flame length in a nonswirling flow is approximately 4% shorter than that in a swirling flow.

This investigation utilized the rifled nozzles with different swirling intensities to increase the interaction between the axial stress of central flow and the axial/tangential stress of swirled annular airflow, and therefore increase the turbulent intensity and the shedding efficiencies between the fuel and swirling jets. The smoke-wire flow visualization, PIV scheme, and topology analysis were applied to observe and analyze the flow structures behind the riffled nozzles. The flame modes, flame height, and flame shedding frequency were also probed.

2 Experiments

2.1 Experimental Setup. Figure 1 schematically depicts the experimental setup. The airflow was driven by a ring-blower at the outlet where an acoustical filter was installed to eliminate the ripples. The central jet comprised of air and C_3H_8 was for flow analysis and flame observations. A pressure regulator, a needle valve, and a rotameter were placed upstream to maintain the flow stability. The experimental apparatus utilized herein were separated into four sections—filter, settling chamber, rifled nozzle, and test sections. In the filter section, the honeycombs and meshes were applied to reduce the flow noise. The honeycombs were used to reduce the radial noise and the meshes installed in parallel were utilized to lower the axial noise. The settling chamber was used to reduce the turbulent intensity and the rifled nozzles were designed to generate swirling flow. In this investigation, three types of nozzles, as listed in Table 1, were adopted. In the test section, the exit diameter of swirling flow (D_o) was 30 mm and the diameter of central flow (d) was 5 mm. A bluff disk (diameter $D=20$ mm) was installed at the exit center. Consequently, the blockage ratio defined by $B=(D/D_o)^2$ was 0.444. The coordinate axes used herein are x and r, which represent the axial and radial axes, respectively. During flame observation, the nozzles burned commercial propane with composition of 95% C_3H_8, 3.5% C_2H_8, and 1.5% C_4H_{10}.

Table 1 Nozzle configurations

Nozzle	Rifles	Outer diameter (mm)	Inner diameter (mm)	Height (mm)	Length (mm)
6-rifled	6	40	30	52	107.6
12-rifled	12	40	30	52	107.6

2.2 Flow Visualization. The flow field was visualized using the smoke-wire visualization. The smoke-streak was produced by heating a zigzag tungsten wire placed close to the jet exit. The diameter of the tungsten wire was 0.2 mm and the zigzag structure was adopted to control the thickness and spacing of the smoke streaks. A thin coating of mineral oil was brushed on the wire surface and then electrically heated to create fine smoke streaks. The Reynolds numbers based on the tungsten-wire diameter were adjusted below 40 to prevent vortex shedding behind the wire [20]. The diameter of aerosol vapor was 1.7 ± 0.3 μm measured by a Malvern 1600C laser particle analyzer. The Stokes number [21] of oil aerosol was at the order of 10^{-3}, which is much lower than one, and therefore the smoke streaks were considered to follow the flow properly. The smoke streaks were illuminated by a 0.6 mm thick laser-light-sheet adjusted by a 20 deg laser-light-sheet expander. The laser wavelength is 540 nm supplied by a Spectra-Physics' Stabilite-2017 6 W argon-ion laser. Two-dimensional streak-images of flow patterns were retrieved using a charge-coupled device (CCD) camera and these images were recorded in a computer at a frame rate of 30 frames/s (fps). The CCD camera is JAI Model CV-A1 with a resolution of 1392×512 pixels. The pixels are 4.65 μm^2 and can be operated continuously or in an external trigger mode. The pixel array is zoomed and mapped to a physical region of approximately 13×5 cm^2 so that the spatial resolution is about 97 μm/pixel. The maximum framing rate of the CCD camera is 30 fps. The Data Translation Mateox Model Meteor-II frame grabber board is employed to digitize the analog voltage output from the CCD array of the camera. During the flame observations, a digital camera (SONY HDR-HC1) was utilized to capture the flame images, which were stored synchronously in a computer. The shutter of the digital camera was set in the range 1/4–1/10,000 s and the frame rate was set at 30 fps.

2.3 PIV Measurement. The PIV image acquisition system, comprised of an argon-ion laser, a high speed camera, and a data translation board, was utilized to measure the velocity contours. The laser light was transmitted by an optical fiber and the thickness of the laser was adjusted by a 20 deg laser-light-sheet ex-

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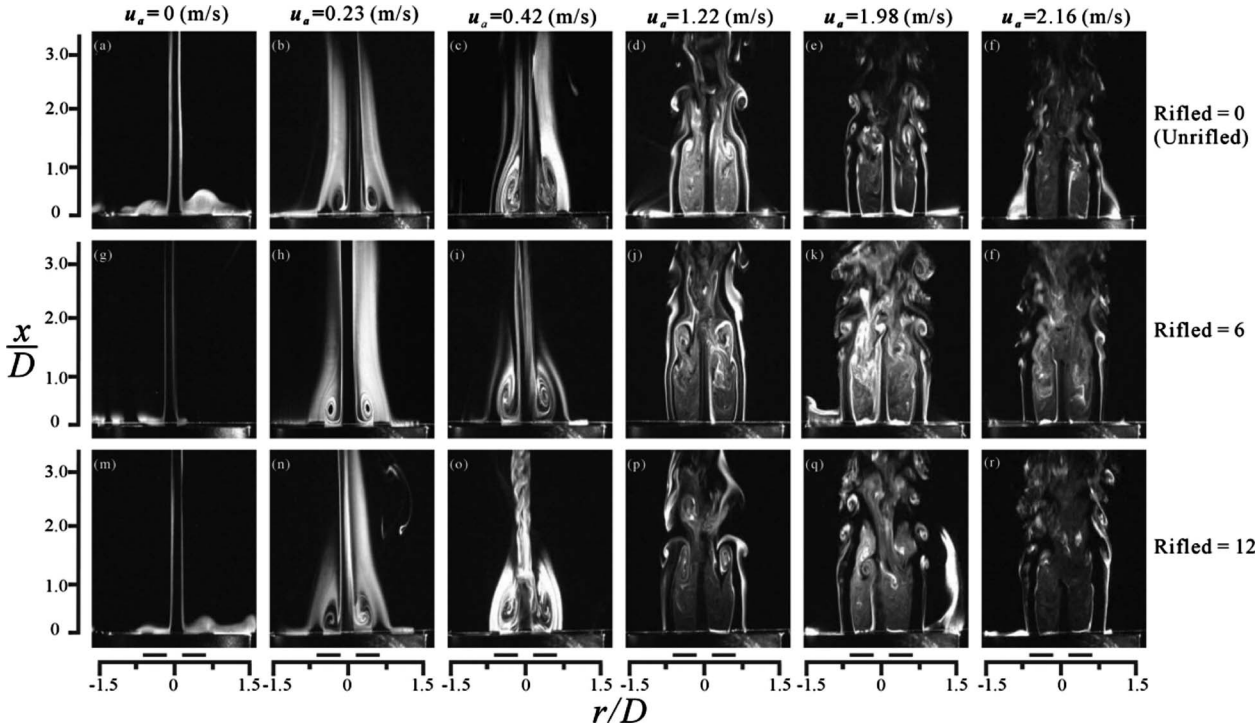

Fig. 2 Streak flow patterns at u_c=2.06 m/s: (*a*)–(*f*) for a unrifled nozzle, (*g*)–(*l*) for a 6-rifled nozzle, and (*m*)–(*r*) for a 12-rifled nozzle

pander. In this study, the thickness of laser was approximately 0.6 mm in the midplane. The wavelength of Ar-ion laser is 540 nm supplied by a Spectra-Physics' Stabilite-2017 6 W argon-ion laser. The particle images were recorded using a CCD camera with a resolution of 480×420 pixels (Redlake MASD, Inc., San Diego, CA, Model MotionScope PCI 2000S). The pixel array of the CCD was zoomed in and mapped onto a physical region of approximately 10×10 cm^2 to yield a spatial resolution of about 208 μm/pixel. The maximum frame rate was 2000 fps and the exposure time was adjustable in the range of 0.025–20 ms. The data translation board (MotionScope PCI controller) was employed to digitize the analog voltage output from the CCD camera. The magnesium oxide (MgO) particles were seeded in the flow field to scatter laser light. The diameter of MgO particles is 5 μm and the density is 3580 kg/m^3 at 25°C. The relaxation time constant is estimated to be less than 279 μs and the Stokes number is 0.007 within the range of experiment. Therefore, the slip between the flow and particles could be neglected.

Abernethy et al. [22] derived the uncertainty (U) in the PIV measurement. According to their results, the 95% confidence levels were formulated as follows:

$$U=[E^2+(tS_{\bar{x}})^2]^{1/2} \tag{1}$$

where E is the bias error, $S_{\bar{x}}$ is the precision index of the average value, and the student's distribution t value equals 2 for a large sample. The bias error is estimated from the measured data or previous experimental judgments and the precision index of the mean is determined from the random error of the measured data [22]. Consequently, the uncertainty of the flow velocity was approximately 1.8% in this work.

2.4 Swirling Number. The swirling number is defined in Eq. (2) [1,16] and specifies the ratio of the axial component of angular momentum flux to the axial component of axial momentum flux.

$$S=\int_{D/2}^{D_o/2} uwr^2dr\Big/\frac{D_m}{2}\int_{D/2}^{D_o/2} u^2rdr \tag{2}$$

where D_m is the mean diameter, that is, $(D+D_o)/2$, u represents the axial velocity and w denotes the tangential velocity near the nozzle exit.

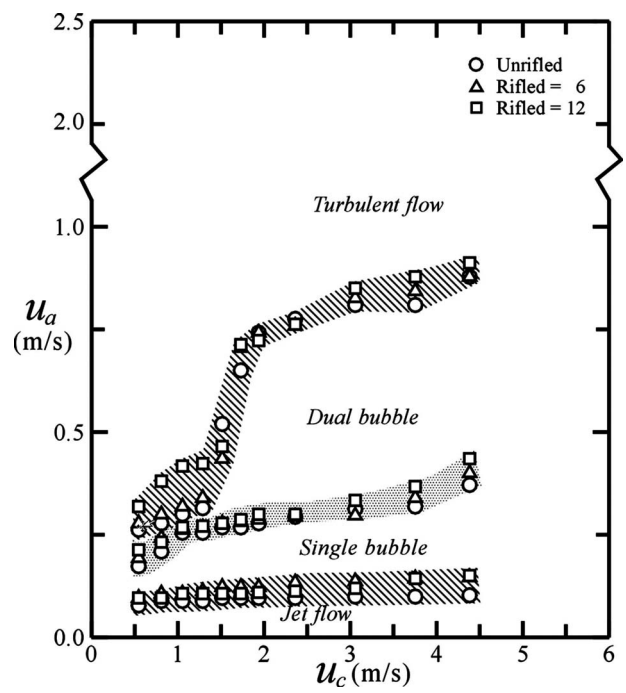

Fig. 3 Distribution of characteristic cold flow modes

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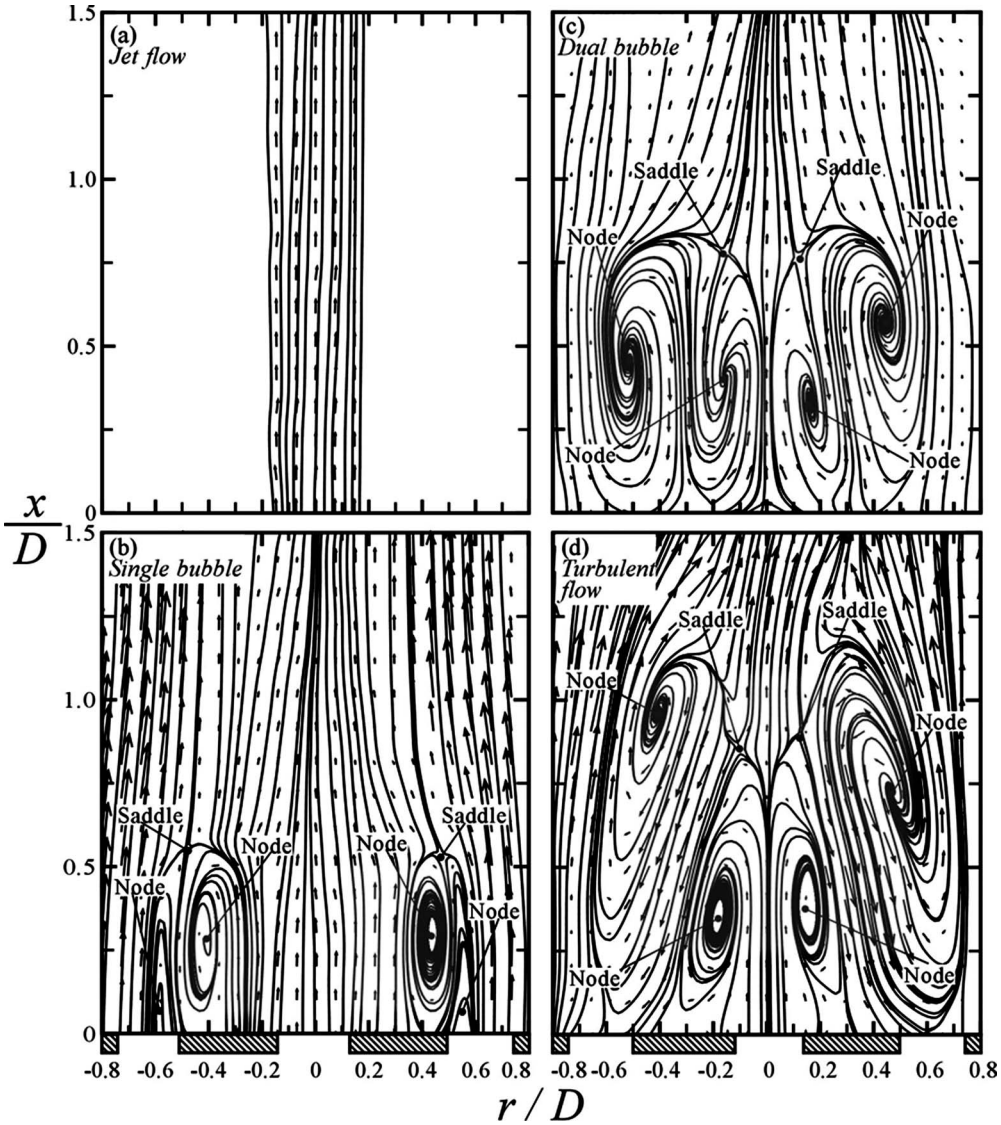

Fig. 4 Velocity vector fields and streamline patterns at u_c=2.06 m/s: (*a*) jet-flow mode (u_a=0), (*b*) single bubble mode (u_a=0.23 m/s), (*c*) dual bubble mode (u_a=0.42 m/s), and (*d*) turbulent flow mode (u_a=1.22 m/s)

2.5 Hot-Wire Anemometry. The vortex shedding behaviors were obtained using a hot-wire probe (TSI 1220-PI2.5) and a fast Fourier transform (FFT) analyzer (CF-920, ONO SOKKI Co., Yokohama, Japan). The highest temperature at which the hot-wire probe can operate was 300°C, and the diameter and length of hot-wire were 5 μm and 1.5 mm, respectively, yielding a dynamic response of 15–25 kHz. The accuracy of the frequency measurement depends both on the response of the hot-wire anemometer and on the record length and sampling rate of the FFT analyzer. The sampling frequency was set at 16 kHz and the uncertainty in the shedding frequency was estimated to be $<3\%$.

3 Results and Discussion

3.1 Smoke-Wire Visualization and Characteristic Flow Patterns. The smoke-wire visualization associated with the laser-light-sheet was employed to observe the recirculation structures of cold flow at different speed ratios of the central jet to the annular jet. Figure 2 shows the visualized smoke-wire patterns of the three nozzles when the central jet speed (u_c) was maintained constant at 2.06 m/s, whereas the annular jet speed (u_a) was varied from 0 m/s to 2.16 m/s.

Figures 2(*a*)–2(*f*) display the flow patterns at various annular jet speeds using a conventional unrifled nozzle. In Fig. 2(*a*), a straight flow that is symmetric about the *x*-axis moves downstream at $u_a=0$ (with no air annular jet) since the central momentum flux moves through the flow field without reverse flow. The flow pattern is defined as the *jet-flow mode*. Figure 2(*b*) shows a pair of stable recirculation bubbles that are produced by the entrainment effect at $u_a=0.23$ m/s. The flow pattern is dominated by the central jet, which can penetrate the recirculation zone due to the high axial momentum of central jet. The flow structure is called the *single bubble mode*. In Fig. 2(*c*), for $u_a=0.42$ m/s, two pairs of dual recirculation bubbles are observed. The inner bubbles drive the outer bubbles to move downstream since the central jet speed exceeds the annular jet speed. Additionally, the height of the recirculation bubble is increased to be higher than that in single bubble mode. The flow pattern is called the *dual bubble mode*. Figure 2(*d*) depicts a recirculation zone at $u_a=1.22$ m/s. The central jet collides with the annular flow at the top of the recirculation zone and an eddy-induced coherent structure moves downstream because of the shear stress effect in the

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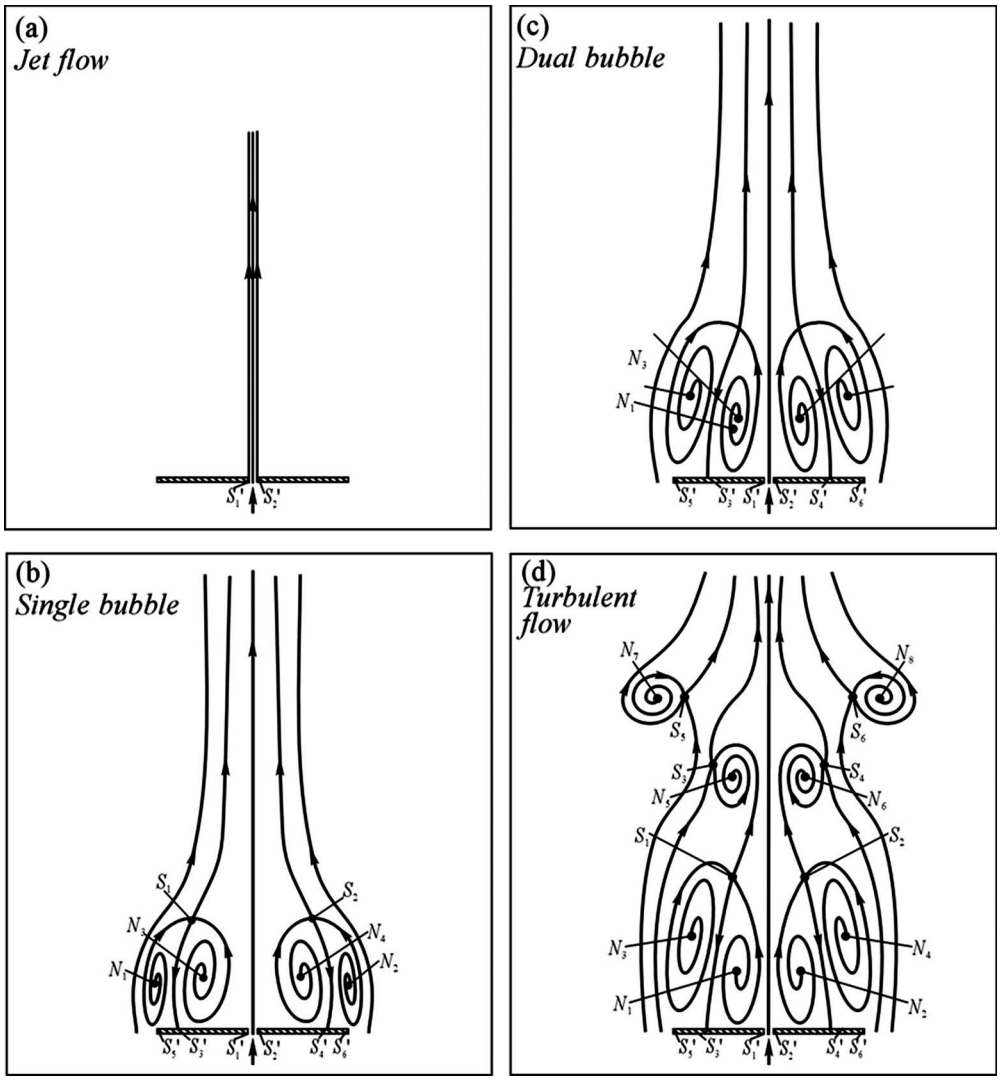

Fig. 5 Topological flow structures at u_c=2.06 m/s: (*a*) jet-flow mode (u_a=0), (*b*) single bubble mode (u_a=0.23 m/s), (*c*) dual bubble mode (u_a=0.42 m/s), and (*d*) turbulent flow mode (u_a =1.22 m/s)

annular flow. Furthermore, the increase in the annular flow velocity increases the bubble height. In Fig. 2(*e*), at u_a=1.98 m/s, the recirculation zone blocks the central momentum flux and a four-way saddle is formed lower than that in Fig. 2(*d*). In Fig. 2(*f*), at u_a=2.16 m/s, the position of the four-way saddle is lower than that in Fig. 2(*e*), and the turbulent intensity in the top region increases with the annular flow velocity. Accordingly, the flow patterns in Figs. 2(*d*)–2(*f*) are associated with the turbulent flow mode.

Figures 2(*g*)–2(*l*) present the flow structures as u_a changes from 0 m/s to 2.16 m/s for a 6-rifled nozzle. Figure 2(*i*) depicts a spiraled central airflow that moves downstream due to the angular acceleration of the rifled nozzle at u_a=0.42 m/s and S=0.15. The outer recirculation is entrained into the inner recirculation and then forms a swirling flow that moves forward along the central axis. In Fig. 2(*l*), when u_a=2.16 and S=0.45, the position of the four-way saddle is lower than that in Fig. 2(*f*) because of the momentum increase in the annular jet. Figures 2(*m*)–2(*r*) display the flow structures for a 12-rifled nozzle at various Reynolds numbers of the annular jet. In Fig. 2(*o*), when u_a=0.42 m/s and S =0.255, the four-way saddle is closer to the bluff disk than those in Figs. 2(*c*) and 2(*i*). Additionally, Fig. 2(*o*) shows a clear spiral jet adjacent to the central axis because of the high swirling strength. In Fig. 2(*r*), when u_a=2.16 m/s and S=0.512, the location of the four-way saddle is lower than those in Figs. 2(*f*) and 2(*l*) due to the increased intensity of the annular jet.

Figure 3 shows the mode regimes in the ranges of $0<u_a<2.5$ m/s and $0<u_c<6$ m/s according to the classification scheme utilized in Fig. 2. The four modes are distinguished clearly at $u_c>2$ m/s because the central airflow velocity is sufficiently high. The regime borders, which separate different characteristic modes are somewhat uncertain. The maximum uncertainty in both u_a and u_c at each mode border is about 0.01 m/s.

3.2 Streamline Vectors. Figure 4 plots the velocity vectors and instantaneous streamline patterns, which were visualized using the PIV scheme. Figure 4(*a*) shows the streamline pattern of the jet-flow mode. Without interference from the annular jet (u_a =0), the central jet streamlines move straight up from the disk center toward the downstream and no recirculation (no node or saddle) occurs in this mode. Figure 4(*b*) shows the streamline pattern of the single bubble mode in which two major recirculation bubbles are observed on both sides of the x-axis and two minor recirculation bubbles, which are absent when the streak-flow pattern occurs adjacent to the disk surface. These two minor recirculation bubbles were formed due to the high intensity in the

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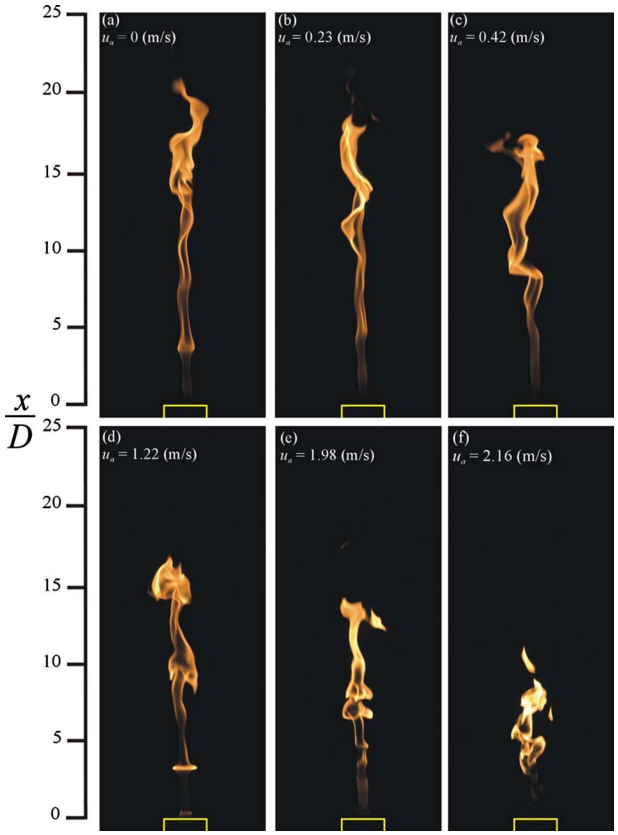

Fig. 6 Photographs of normalized flame height (x/D) for a unrifled nozzle at u_c=2.06 m/s: (*a*) u_a=0, (*b*) u_a=0.23, (*c*) u_a =0.42, (*d*) u_a=1.22, (*e*) u_a=1.98, and (*f*) u_a=2.16 m/s

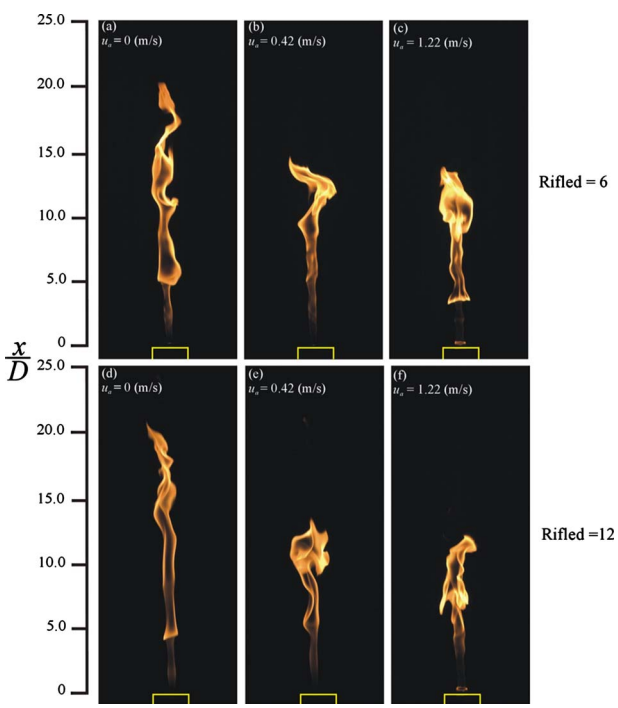

Fig. 7 Photographs of normalized flame height (x/D) at u_c =2.06 m/s: (*a*)–(*c*) for a 6-rifled nozzle and (*d*)–(*f*) for a 12-rifled nozzle

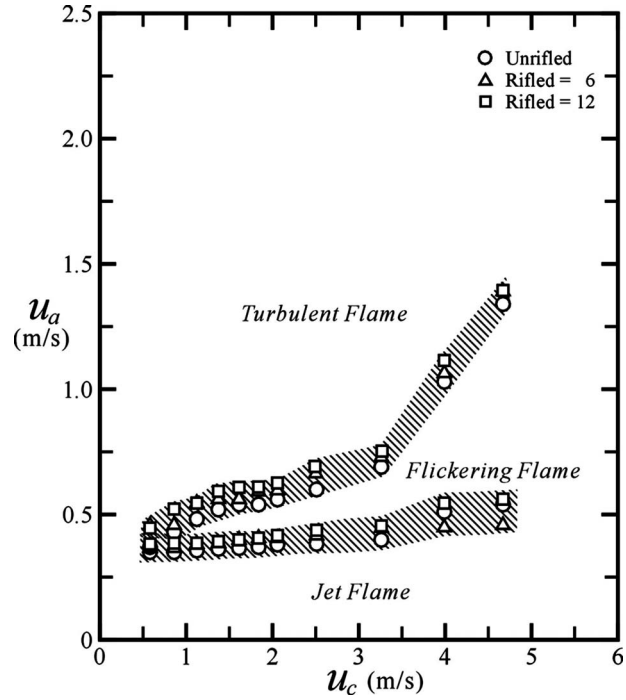

Fig. 8 Distribution of characteristic flame configurations

major bubbles, such that the annular jet separates and is reversed due to the entrainment effect. Moreover, the normalized height of recirculation zone (x/D) is about 0.55. Figure 4(*c*) shows the streamline pattern of dual bubble mode. Two pairs of dual bubbles were formed symmetrically about the x-axis. The normalized height of the outer bubbles (x/D) is about 0.80. Accordingly, the bubble scale in Fig. 4(*c*) is larger than that in Fig. 4(*b*). Figure 4(*d*) shows the streamline pattern of the turbulent flow mode where two pairs of dual bubbles are observed on each side of the x-axis. The normalized height (x/D) of the outer bubbles is $\approx$1.15. The height of the outer bubbles is extended because the momentum in annular airflow is sufficiently high.

3.3 Topological Analysis. Perry and Fairlie [23], Chong and Perry [24], and Lighthill [25] developed the topological sketch

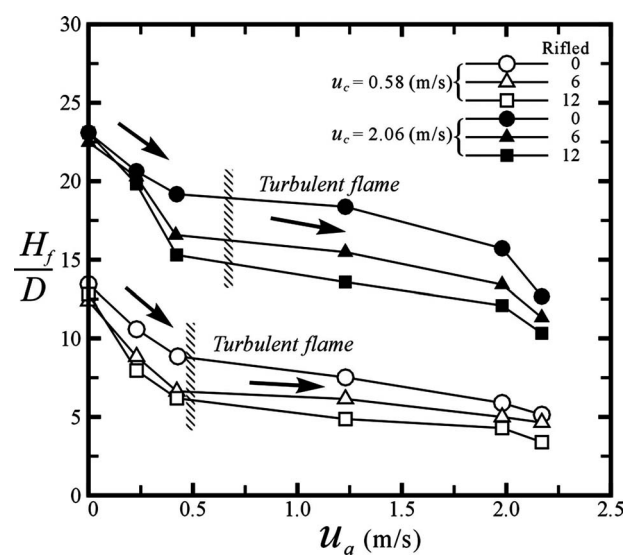

Fig. 9 Normalized flame height (H_f/D) against u_a at u_c =0.58 m/s (void symbols) and u_c=2.06 m/s (solid symbols)

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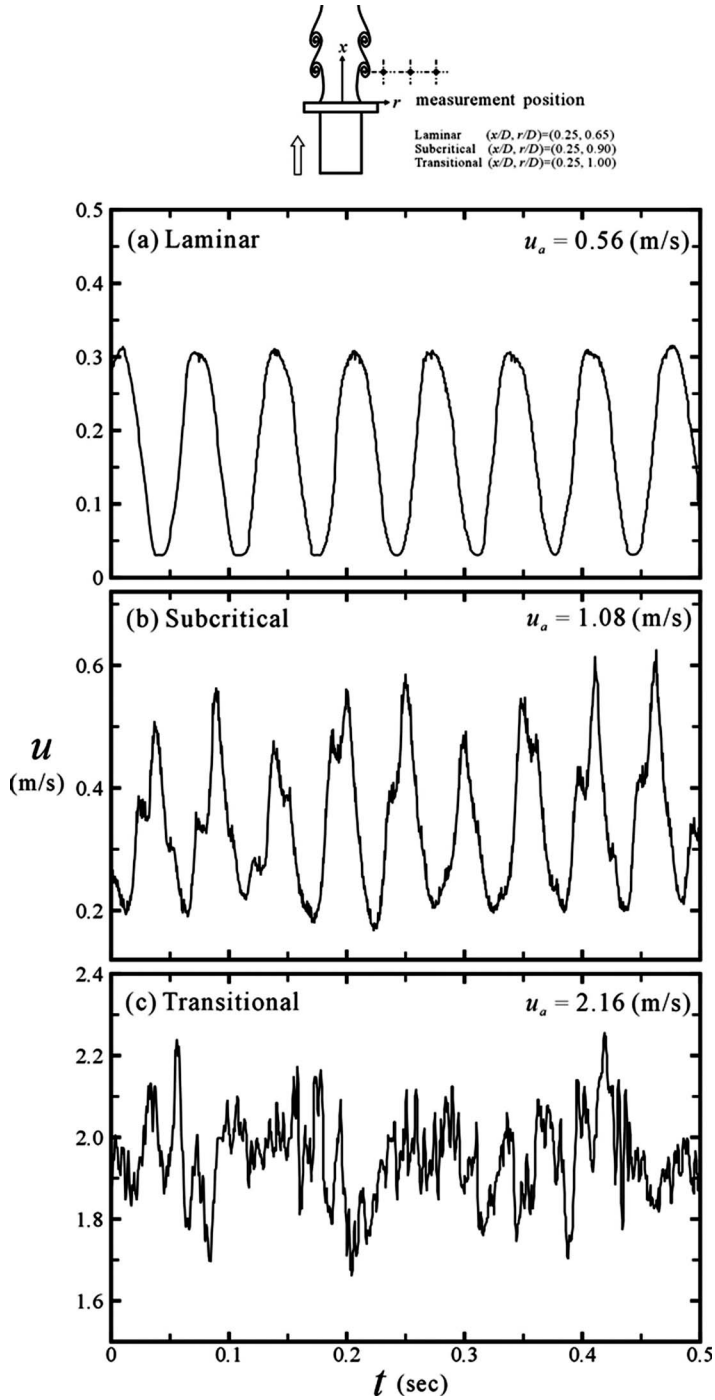

Fig. 10 Velocity profiles of flame shedding for (*a*) laminar, (*b*) subcritical, and (*c*) transitional modes

scheme to analyze the flow patterns. Perry and Steiner [26] adopted the critical point theory to determine the sectional streamline patterns and defined two main types of singular point—nodes and saddles. A node is associated with either attachment or separation. The topological lines are directed away from a nodal point of attachment but toward a nodal point of separation. Only two topological lines pass through a saddle point. One is inward and the other is outward [27]. For flow around a three-dimensional entity, some singular points in the flow pattern may be formed at particular positions where the velocities are zero and the streamline slope is indeterminate [28]. The topological scheme can reveal the steady flow structures and the evolution of the unsteady vortical structures. Finally, Hunt et al. [29] utilized the topological theory to formulate the relationship between nodes (including the four-way node N and the three-way node N'), saddles (including the four-way saddle S and the three-way saddle S') and obstacles in the flow field as follows:

$$\left(\sum N + \frac{1}{2}\sum N'\right) - \left(\sum S + \frac{1}{2}\sum S'\right) = 1 - n \qquad (3)$$

where n is the connectivity of the flow section. For instance, $n=1$ for a single-connected region without obstacle, $n=2$ for one

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entity, and $n=3$ for two entities present in the flow field.

Figure 5 plots the topological flow configurations of the four characteristic modes in the cold flow field. In Fig. 5(*a*), the topological configuration of the jet-flow mode reveals that the values of the critical points are $\Sigma N=0$, $\Sigma N'=0$, $\Sigma S=0$, and $\Sigma S'=2$, which fulfill the topological condition (Eq. (3)). Figure 5(*b*) depicts the topological flow configuration of the single bubble mode and four nodes (N_1-N_4): two four-way saddles (S_1 and S_2) and six three-way saddles ($S_1'-S_6'$). Accordingly, the values of critical points are $\Sigma N=4$, $\Sigma N'=0$, $\Sigma S=2$, and $\Sigma S'=6$, which satisfy Eq. (3). Figure 5(*c*) displays the topological flow configuration of the dual bubble mode. The values of critical points are the same as those determined in the single bubble mode. Accordingly, the topological condition is also satisfied in the dual bubble mode. Figure 5(*d*) presents the topological configuration in the turbulent flow mode where eight nodes (N_1-N_8), four four-way saddles (S_1-S_4), and six three-way saddles ($S_1'-S_6'$) are identified, and fulfill the topological condition.

3.4 Direct-Photo Visualization and Flame Modes. The fuel jet was discharged from the center of a bluff disk and was mixed with the annular air jet behind the unrifled, 6-rifled, and 12-rifled nozzles. The mixed flow was ignited behind the bluff disk, while the velocities of both the central fuel jet and the annular airflow were varied to observe directly the flame shapes. Figures 6 and 7 display the photographs of the flame configurations behind the nozzles.

For an unrifled nozzle, the central jet velocity is maintained at $u_c=2.06$ m/s and the annular airflow velocity is changed from 0 m/s to 2.16 m/s. Figure 6(*a*) shows the flame height without interference from annular airflow (i.e., $u_a=0$). Figure 6(*b*) shows the flame configuration at $u_a=0.23$ m/s. The flame structure is assembled as displayed in Fig. 6(*a*) because of the weak pressure gradient behind the bluff disk. The flow structures in Figs. 6(*a*) and 6(*b*) are those of the *jet flame mode*. Figure 6(*c*) shows an unstable flickering flame at $u_a=0.42$ m/s, and the flame structure is denoted as the *flickering flame mode*.

The flame height in Fig. 6(*d*) is shorter than that in Fig. 6(*a*) at $u_a=1.22$ m/s. The normalized flame height (x/D) of the recirculation bubble is approximately 0.8. Figure 6(*e*) shows a stable ring flame occurring at $x/D\approx 0.5$ at $u_a=1.98$ m/s. Figure 6(*f*) indicates that the flame is lifted at $x/D\approx 1.2$ when $u_a=2.16$ m/s. The flame structures in Figs. 6(*d*)–6(*f*) are called the *turbulent flame modes*.

3.4.1 Characteristic Flame Modes in Swirled Wake. Figure 7 depicts the characteristic flame configurations in the swirled wake behind the rifled nozzles. Figures 7(*a*)–7(*c*) present the modulated flame structures associated with a 6-rifled nozzle and Figs. 7(*d*)–7(*f*) present the flame configurations for a 12-rifled nozzle with a constant u_c of 2.06 m/s. Figures 7(*a*) and 7(*d*) show the modulated flame structures without interference from the annular swirled airflow ($u_a=0$). The flame structures are similar to those shown in Figs. 6(*a*) and 6(*b*), and therefore are called the jet flame modes. Figures 7(*b*) and 7(*e*) present an unstable flickering flame at $u_a=0.42$ m/s. Consequently, the flame structures are named the *flickering flame mode*. Figures 7(*c*) and 7(*f*) present the recirculation zones in the range of $x/D<0.8$ at $u_a=1.22$ m/s. The flame structures are then called the turbulent flame modes.

Figure 8 shows the distributions of the three flame modes as u_a and u_c varied from 0 m/s to 2.5 m/s and from 0 m/s to 6 m/s, respectively. The regime borders, which separate different characteristic flame modes, are associated with some uncertainties. The maximum uncertainty for u_a and u_c is 0.01 m/s.

3.4.2 Flame Height. Figure 9 shows the profiles of the normalized flame height (H_f/D) as u_a is varied from 0 m/s to 2.5 m/s at $u_c=0.58$ m/s and 2.06 m/s for the three nozzles utilized herein. The flame height in this manuscript was measured by the traditional photography scheme by using a Nikon Model D70s CCD with a resolution of 3008×2000 pixels and an exposure time of 30–1/8000 s. The flame height in this study was defined by averaging ten taken photos with an exposure time of 2 s for each photo. Figure 9 indicates that H_f/D declines as u_a increases. The flame height for $u_c=2.06$ m/s exceeds that for $u_c=0.58$ m/s at the same u_a because more propane is released. The flame heights behind the 12-rifled nozzle are lower than those of the unrifled and 6-rifled nozzles, since more swirling airflow is generated. Additionally, the flame height of a 12-rifled nozzle is approximately halved with comparing that for an unrifled nozzles. The swirling effect changes the axial jet flow to a radial one in the tangential direction. The swirling mechanism creates significant momentum, heat exchange, mass exchange, and drastic turbulence fluctuations. Consequently, the evident increases in the swirling intensity shorten the flame length and thus increase combustion efficiency.

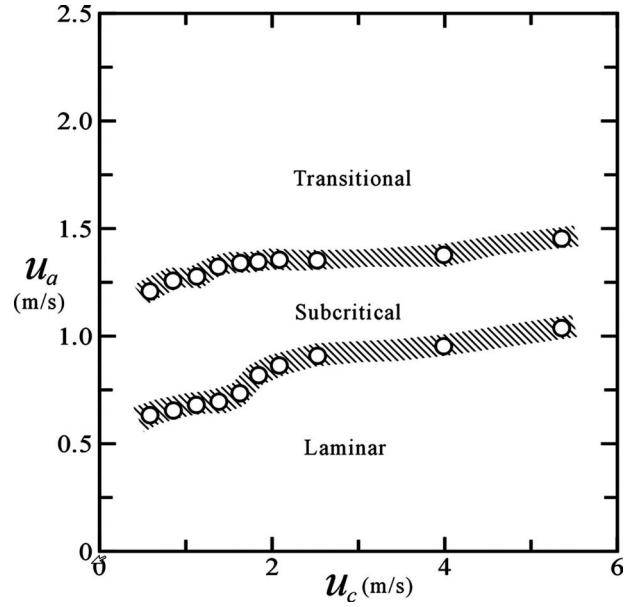

Fig. 11 Distribution of flame shedding modes

3.5 Vortex Shedding of Flame. The vortex shedding of the flame behind the nozzles was probed using a hot-wire anemometer. Figure 10 shows the velocity profiles (u) for various evolution times (t) at $u_a=0.56$ m/s, 1.08 m/s, and 2.15 m/s, while u_c is fixed at 2.06 m/s. To collect clear signals, a hot-wire probe was installed at $(x/D, r/D)=(0.25, 0.65)$, (0.25,0.90), and (0.25, 1.00) in the laminar, subcritical, and transitional modes, respectively. Figure 10(*a*) indicates that the time signals are periodic (sinusoidal wave) at $u_a=0.56$ m/s. Such vortex shedding is associated with the *laminar mode*. Figure 10(*b*) shows periodic signals at $u_a=1.08$ m/s, but some weak disturbances are multiplied in the sinusoidal waves. Such vortex shedding is named the *subcritical mode*. Figure 10(*c*) presents a random output at $u_a=2.16$ m/s, and therefore the vortex shedding profile is called the *transitional mode*. Figure 11 shows the distribution of vortex shedding modes as u_a and u_c vary in the ranges 0–2.5 m/s and 0–6 m/s, respectively.

Figure 12 displays the spectrum ($\log \Phi$) calculated using the FFT algorithm for the three vortex shedding modes, where Φ represents the power spectrum. Figure 12(*a*) shows a peak frequency of 11.90 Hz at $u_a=0.56$ m/s and $u_c=2.06$ m/s (i.e., in laminar vortex shedding mode). Figure 12(*b*) shows a peak frequency of 16.05 Hz at $u_a=1.08$ m/s (i.e., in subcritical vortex shedding mode). Figure 12(*c*) shows no peak frequency in the transitional vortex shedding mode at $u_a=2.16$ m/s. Consequently,

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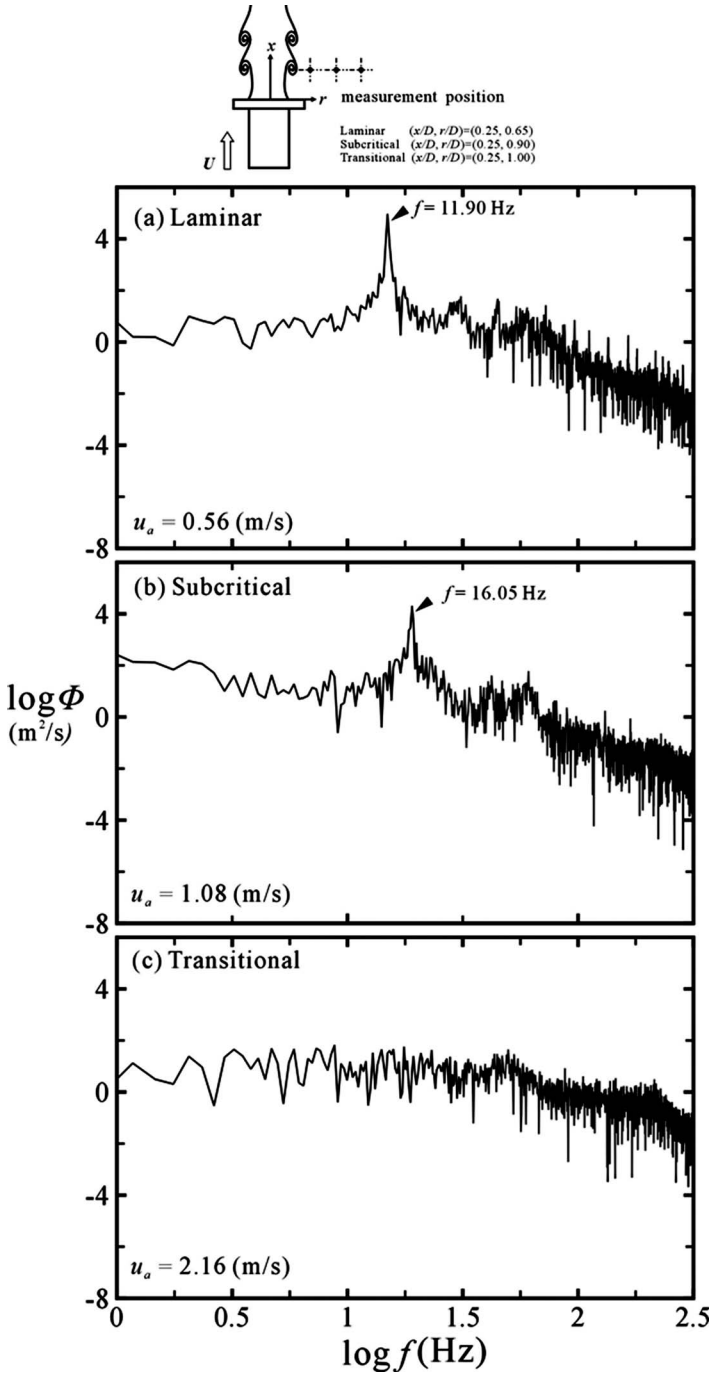

Fig. 12 Distributions of power spectrum (log Φ) versus logarithmic frequency (log *f*) for flame shedding in (*a*) laminar, (*b*) subcritical, and (c) transitional modes

no coherent flame structure is observed in the transitional mode.

Figure 13 shows the distributions of the pdf derived from the data given in Fig. 10. Figure 13(*a*) shows a dual-peak pdf in the laminar vortex shedding mode that is associated with the sinusoidal output plotted in Fig. 10(*a*). Figure 13(*b*) shows a Gaussian-like distribution in the subcritical vortex shedding mode associated with the multiplication of weak disturbance in the output signals. Figure 13(*c*) plots Gaussian distributions due to the random fluctuations of shedding velocities in the transitional vortex shedding mode.

The frequency of unsteady flow in the wake varying with free velocity is normalized and represented by the nondimensional groups: Strouhal number (Sr) is defined by $\mathrm{Sr}=fD/u_\infty$. Roshko [30] experimentally developed a well-known equation for the nondimensional frequency Sr of the Kármán vortex shedding in the wake behind a circular cylinder, as follows:

$$\mathrm{Sr} = 0.212 - \frac{4.5}{\mathrm{Re}} \quad \text{for } 300 < \mathrm{Re} < 2000 \tag{4}$$

Therefore, the limit of Strouhal number for a circular cylinder is 0.212. Figure 14 displays the distributions of the shedding frequency (f) and the Strouhal number with changing u_a in the range 0–1.6 m/s at two u_c of 0.58 m/s and 2.06 m/s. Figure 14(*a*) iden-

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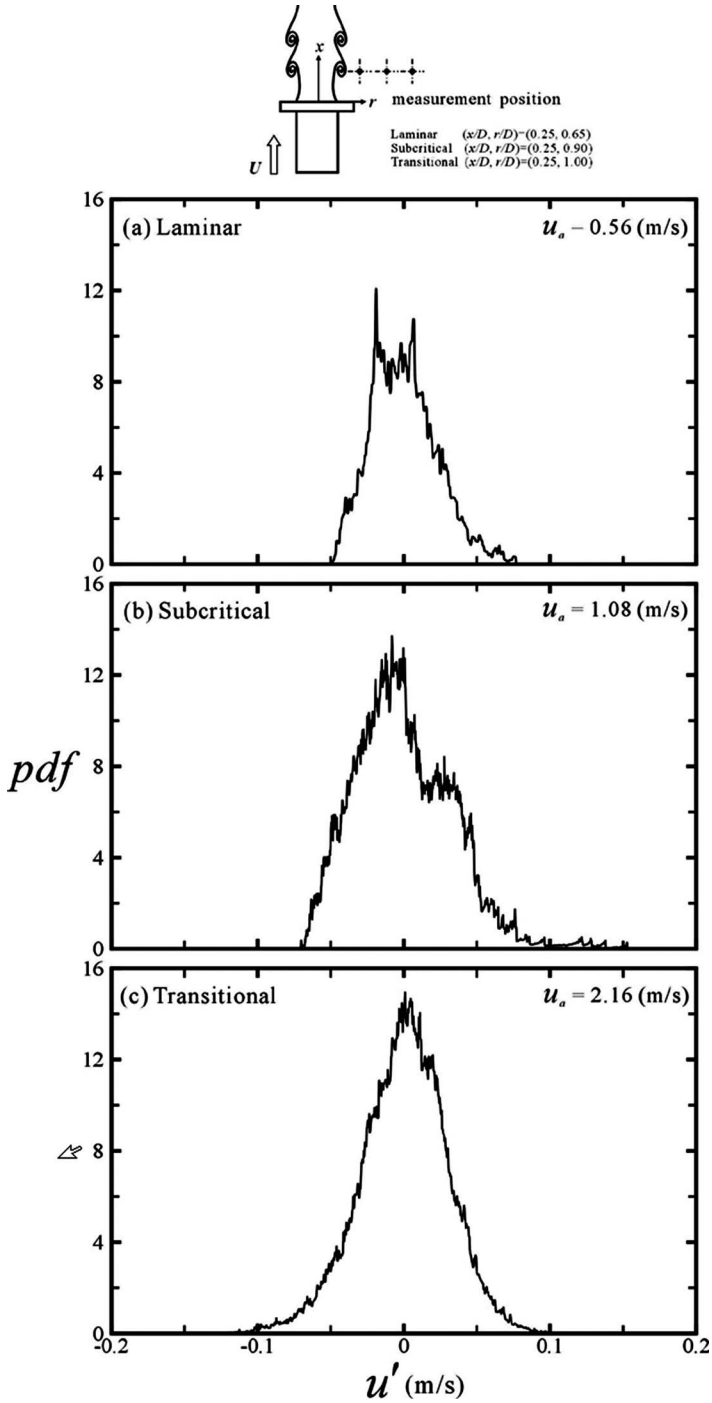

Fig. 13 Distributions of probability density function as a function of u' for flame shedding in (*a*) laminar, (*b*) subcritical, and (*c*) transitional modes

tifies three flame regimes—jet flame, flickering flame, and turbulent flame. In the jet flame regime, the shedding frequency increases with u_a at $u_a < 0.42$ m/s. In the flickering flame regime, the shedding frequency declines rapidly as u_a increases in the range of $0.42 < u_a < 0.5$ m/s. In the turbulent flame regime, the shedding frequency increases with u_a at $u_a > 0.5$ m/s. Figure 14(*b*) shows the profile of Sr against u_a, where Sr is defined as $\mathrm{Sr} = fd/u_a$. The Sr declines as u_a increases, and the relationship between Sr and u_a can be fitted using the following first-order correlation:

$$\mathrm{Sr} = 0.0238 + 0.13/u_a \tag{5}$$

4 Conclusion

This work adopted rifled nozzles with various swirl numbers to increase interaction between the axial stress of the central flow and the tangential stress of the swirled annular airflow. Additionally, the rifled nozzles are designed to increase the turbulent intensity and thereby increase the mixing and shedding efficiencies between the central flow and the swirled annular airflow. Smoke-wire flow visualization, the PIV scheme, topological analysis, and direct-photo visualization were adopted to observe and analyze the cold flow structures and flame configurations behind the riffled

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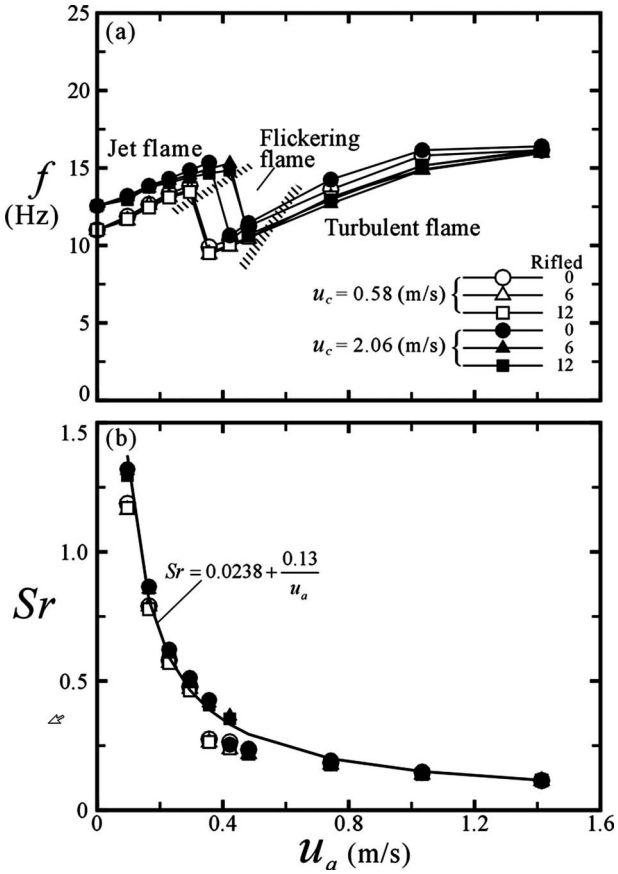

Fig. 14 Distributions of (*a*) *f* versus u_a and (*b*) Sr versus u_a

nozzles. The experimental data and discussion yield the following results.

(1) The cold flow structures behind the rifled/unrifled nozzles are classified into four modes—jet flow, single bubble, dual bubble, and turbulent flow—based on smoke-wire visualization.
(2) The flame configurations behind the rifled/unrifled nozzles are classified into three modes—jet flame, flickering flame, and turbulent flame—based on direct-photo visualization.
(3) The flame height declines as u_a increases, and the flame height behind a 12-rifled nozzle is decreased by about 50% under that of an unrifled nozzle because of the sufficiently high swirling intensity.
(4) The flame shedding frequency behind the nozzles declines rapidly in the flickering flame mode, and the relationship between Sr and u_a is regressed as $\mathrm{Sr}=0.0238+0.13/u_a$.

Nomenclature

A_a = exit area of swirling jet: $\pi(D_o^2\text{-}D^2)/4$, 392.7 mm^2
A_c = exit area of central jet: $\pi d^2/4$, 19.6 mm^2
B = blockage ratio at jet exit: D^2/D_o^2, 0.444
D = diameter of blockage disk, 20 mm
d = exit diameter of central jet, 5 mm
D_c = diameter of control disk, 10 mm
D_h = hydraulic diameter of annular swirling jet at exit: $D_o\text{-}D$, 10 mm
D_m = mean diameter for calculation of swirl number: $(D+D_o)/2$, 25 mm
D_o = outer diameter of annular swirling jet at exit, 30 mm
f = frequency of instabilities in wake (Hz)
Q_a = volumetric flow rate of annular swirling flow
Q_c = volumetric flow rate of central jet
r = radial coordinate, originated at center of blockage disk
Ro = Roshko number of vortex shedding: fd^2/ν
Sr = Strouhal number of vortex shedding: fd/u
u = axial velocity
x = axial coordinate, originated at the center of the blockage disk
u_a = volumetric mean axial velocity of annular swirling jet at exit: Q_a/A_a
u_c = volumetric mean velocity of central jet at exit: Q_c/A_c
v = radial velocity
w = azimuthal velocity
Φ = power spectrum of fluctuating velocities
τ = time shift for calculating autocorrelation coefficient of fluctuation velocities

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